Anomaly detection using surprisals

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Abstract

TBC

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1 Introduction

Anomaly detection methods are often rather ad hoc and lack a solid theoretical foundation, or they make strong distributional assumptions that may not hold in practice. We aim to address this by defining an anomaly as a point of low probability, and proposing a general framework for anomaly detection by estimating the probability of observations with lower density (thus more anomalous). Our approach allows for a unified treatment of anomaly detection that can be applied to any anomaly detection problem, provided a probability distribution can be defined on the space of observations. It allows for univariate and multivariate settings, and can be applied to both unconditional and conditional distributions. Although it requires a specified probability distribution, it is robust to substantial mis-specification of that distribution.

The basic idea is to calculate the surprisal (equal to the negative log density) of each observation, and then estimate the tail probability from the surprisal distribution. This allows us to estimate the probability of observing a surprisal value as extreme or more extreme than a given observation, which can then be used to flag anomalies. The surprisal approach to anomaly detection provides a principled way to control the false positive rate when flagging anomalies, by setting a threshold on the estimated tail probabilities.

Let $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ be iid observations from a probability distribution F defined on an appropriate sample space, and let f be the generalized probability density function of F. That is, f is defined using Dirac delta functions at jump points of F, so that it represents the density function if F is continuous, the probability mass function if F is discrete, and a combination of both if F is a mixture of continuous and discrete components.

The *surprisal* of observation $\boldsymbol{y}_i$ is defined as (Stone 2022)

$$s_i = -\log f(\boldsymbol{y}_i).$$

While these are better known as "log scores" in statistics (Good 1952), in this context we prefer the term "surprisal" (coined by Tribus 1961), as it captures the idea that observations with low density are more "surprising" (and therefore potentially anomalous) than those with high density. A large value of s_i indicates that $\boldsymbol{y}_i$ is an unlikely value, and so is a potential anomaly. On the other hand, typical values will have low surprisals. In information theory, the average surprisal is known as the entropy of a random variable (Cover & Thomas 2006; Stone 2022) and provides a measure of uncertainty. If f is obtained from a parametric model, then the sum of the surprisals is equal to minus the log likelihood of the data.

Let $G(s) = \Pr\{S \leq s\}$ be the distribution function of the surprisal values, where $S = -\log f(\boldsymbol{Y})$ and $\boldsymbol{Y} \sim F$. Then the probability of observing a value with surprisal larger than s is given by $1 - G(s)$, which leads to a natural way of assigning an anomaly score to each observation $\boldsymbol{y}_i$:

$$p_i = 1 - G(s_i) = 1 - G(-\log f(\boldsymbol{y}_i)).$$

We define anomalies as observations with $p_i < \alpha$, giving a false positive rate of α if all observations are from F. This is preferable to directly computing the probabilities from the tail of F, as it allows for anomalies to occur anywhere in the distribution of F, not just in its tails. For example, anomalies may lie between modes in a multimodal distribution, or in otherwise low-density regions of the distribution F. It is also preferable to a distance-based approach to anomaly detection (e.g., Wilkinson 2018), as it naturally handles skewed and heavy-tailed distributions.

The surprisal approach converts the problem of working with F, a potentially multivariate distribution (or even one on a manifold or some non-Euclidean space), to working with a univariate distribution G defined on $\mathbb{R}$. We can write

$$G(s) = \Pr\{-\log f(\boldsymbol{Y}) \leq s\} = \Pr\{f(\boldsymbol{Y}) \geq e^{-s}\}.$$

This makes clear that the surprisal probability is related to the highest density regions of the distribution (Hyndman 1996). In fact, every observation with $p_i < \alpha$ lies outside the $100(1-\alpha)\%$ highest density region of F.

If $F = N(\mu, \sigma^2)$, then $p_i = 2[1 - \Phi(|y_i - \mu|/\sigma)]$, where Φ is the standard Normal distribution function. In this case, identifying anomalies as observations with $p_i < \alpha$ is equivalent to a z-score test with the threshold set to $\Phi^{-1}(1 - \alpha/2)$. Hence, our approach can be seen as a generalization of anomaly detection methods based on z-scores including Grubbs' test (Grubbs 1950).

It can also be seen as a generalization of other anomaly detection methods. For example, the Hampel identifier, due to a proposal of Frank Hampel (Davies & Gather 1993), is designed to identify anomalies in a time series based on whether an observation is very different from the neighboring observations. It denotes an observation as an anomaly if $|y_t - m_t|/a_t > \tau$, where

$$\begin{aligned} m_t &= \text{median}(y_{t-h}, \dots, y_{t+h}), \\ \text{and} \qquad a_t &= \text{median}(|y_{t-h} - m_t|, \dots, |y_{t+h} - m_t|). \end{aligned}$$

The threshold τ is often set under the assumption of a Normal distribution. When we take a surprisal perspective, this is equivalent to setting

$$F = N(m_t, a_t/\Phi^{-1}(0.75)).$$

The false anomaly rate is $\alpha = 2[1 - \Phi(\tau\Phi(0.75))]$.

We will usually not know F, and so we need to estimate it from data or from theory. Such estimates are particularly fragile or uncertain in areas of low density, which is precisely where we are looking for anomalies. Further, we are seeking a method that does not make strong distributional assumptions.

Our approach begins with an assumed F, which may be estimated from data, or obtained from theory. This distribution is used to compute the surprisal values $\{s_1, \dots, s_n\}$. Then, we estimate the tail probabilities $\{p_1, \dots, p_n\}$ from the surprisal values, using one of three approaches.

(1) Using the assumed distribution F to compute the tail probabilities, i.e., $p_i = \int I\{f(\boldsymbol{y}) \le e^{-s_i}\} d\boldsymbol{y}$, where I is the indicator function;
(2) Using the empirical distribution function of the observed surprisals $\{s_1, \dots, s_n\}$, so that p_i is the proportion of surprisals larger than s_i;
(3) Fitting a Generalized Pareto Distribution (GPD) to the largest surprisal values, and using this to estimate the tail probabilities.

Obviously approach (1) makes the strong assumption that F is an accurate representation of the data. Approaches (2) and (3) allow for considerable mis-specification of the distribution used to compute the surprisals. Approach (2) requires only that the ordering of the surprisals in the tail is similar to what would have been obtained under the true distribution, while approach (3) requires that extreme value theory applies to the surprisal distribution.

In Section 2, we develop the necessary theory to support the use of empirical surprisal probabilities, and identify the conditions under which this will give a good estimate of the true surprisal probabilities.

Similarly, Section 3 provides the theoretical results needed to use a Generalized Pareto Distribution to approximate the tail of the surprisal distribution. Then we demonstrate the methods with several simulation experiments in Section 4, and in two applications in Section 5. A discussion follows in Section 6. The appendix contains the proofs needed for the results in Sections 2 and 3.

2 Empirical surprisal probabilities

Under approach (2) above, we estimate $p_i = \Pr[S > s_i]$ as the proportion of observed surprisals larger than s_i (equal to $1 - \text{rank}(s_i)/n$). Using this approach, the anomaly boundary for a threshold of α corresponds to the $100(1-\alpha)\%$ highest density region computed using the quantile density approach of Hyndman (1996).

Let $F(\cdot)$ denote the true distribution function of $\boldsymbol{Y}$, and suppose we use $\widehat{F}$ to model $\boldsymbol{Y}$. Their generalized density functions will be denoted by $f(\cdot)$ and $\widehat{f}(\cdot)$ respectively. Then, for this approach to be accurate, the surprisal ranks under $f(\cdot)$ must behave similarly to those obtained under $\widehat{f}$. In particular, since we are only interested in large surprisal values, *it is both necessary and sufficient* that the surprisal ranks in the upper tail of the surprisal distribution can be related through a monotone transformation.

Recall that the surprisal under the true distribution is defined as $S = -\log f(\boldsymbol{Y})$, and recall that $G(s) = \Pr(S \le s)$ denotes the CDF of S at $s \in \mathbb{S} \subseteq \mathbb{R}$, where $\mathbb{S}$ denotes the support of the surprisals. Likewise, let $\widehat{S} = -\log \widehat{f}(\boldsymbol{Y})$ denote the surprisal under the fitted distribution, and let $\widehat{\mathbb{P}}_n(s) := n^{-1}\sum_{i=1}^{n} 1[\widehat{S}_i \le s]$ denote the empirical CDF (eCDF) of the fitted sample $\widehat{S}_1, \dots, \widehat{S}_n$. For $\widehat{\mathbb{P}}_n(s)$ to be an accurate estimate of $G(s)$ in any given sample, the following condition turns out to be both necessary and sufficient.

Assumption 2.1. There exists an $s_\star$ such that, for all $s \ge s_\star$ and a function $h(\cdot)$, strictly increasing on $s \ge s_\star$, such that $S = h(\widehat{S})$ almost surely.

Lemma 2.1. *Assume that $S_1, \dots, S_n$ are iid realizations from G. For $\alpha \in (0,1)$, with probability at least $1-\alpha$, for any $s \ge s_\star$, $\widehat{\mathbb{P}}_n(s) - \epsilon \le G(s) \le \widehat{\mathbb{P}}_n + \epsilon$ if and only if Assumption 2.1 is satisfied.*

Check notation

Lemma 2.1 implies that if the fitted distribution has tails that are similarly located to the actual data distribution, then the fitted empirical surprisal probabilities are very accurate measures of the true surprisal probability, $G(s)$. Critically, this does not require that the tails of the fitted distribution are the same as the actual data distribution, and only requires that they be related through a monotone transformation. That is, for this approach to be accurate, we must be confident that the fitted distribution has regions of low density in the same locations as the true distribution.

Importantly, Assumption 2.1 is not satisfied by all pairs of distributions: if the true data has skewness but the fitted model assumes symmetry, or if there is bimodality in the observed data that is not modeled, then Assumption 2.1 will not necessarily be satisfied. We forego a detailed example of the success and failure of this approach until Section ???.

3 Extreme value theory for the surprisal distribution

Our third approach uses extreme value theory applied to the surprisal values. We will show that under weak conditions on the tail of the surprisal distribution, the maximum surprisal converges to one of the three types of extreme value distributions. This justifies fitting a Generalized Pareto Distribution (GPD) to the largest surprisal values, and using it to estimate tail probabilities.

An important precursor to this work is Kandanaarachchi & Hyndman (2022) who proposed estimating F by a leave-one-out kernel density estimate, and then fitting a Generalized Pareto Distribution (GPD) to the most extreme surprisal values. The underlying idea is that even if F is not particularly well-estimated, the tail of the surprisal distribution should still be well-approximated by a GPD. Hyndman, Kandanaarachchi & Turner (2025) provides some theoretical justification for this approach in the context of kernel density estimation.

Here we extend that idea to allow F to be any well-behaved distribution, including conditional distributions. Specifically, we fit a GPD to the largest β of $\{s_i\}_{i=1}^n$, obtaining estimates of the location, scale and shape parameters, (μ, σ, ξ). We calculate the probability of a surprisal more extreme than each observation, $p_i = \beta\left[1 - P(s_i \mid \hat{\mu}, \hat{\sigma}, \hat{\xi})\right]$, where P is the distribution function of the GPD. Observations not contained in the largest β of $\{s_i\}_{i=1}^n$ are assigned $p_i = \beta$. To justify this approach, we seek conditions on the surprisal distribution of $S = -\log f(\boldsymbol{Y})$, where $\boldsymbol{Y} \sim F$, and f is corresponding generalized density function, similar to the Fisher-Tippett-Gnedenko (FTG) theorem (Coles 2001).

We consider three possible conditions on the surprisal distribution of $S = -\log f(\boldsymbol{Y})$, describing increasingly heavy tails:

1. where S is sub-Gaussian (e.g., when f is bounded above and below);
2. where S is sub-exponential (e.g., when f is Gaussian); and
3. where S has a polynomial moment (e.g., when f is Student-t).

We state these as three alternative assumptions.

Assumption 3.1 (Sub-Gaussian)**.** The random variable $S = -\log f(\boldsymbol{Y}) \in \mathbb{R}$, and satisfies, for all $\lambda \in \mathbb{R}$, and some $\nu > 0$,

$$\mathbb{E}\exp\{\lambda(S - \mathbb{E}[S])\} \leq \exp\{\lambda^2 \nu^2/2\}.$$

Assumption 3.2 (Sub-exponential)**.** The random variable S is sub-exponential with parameters ν and b, i.e.,

$$\mathbb{E}\exp\{\lambda(S-\mathbb{E}[S])\} \leq \exp\{\lambda^2\nu^2/2\} \quad \text{for all} \quad |\lambda| < 1/b.$$

Assumption 3.3 (Polynomial)**.** The random variable $|S|$ has polynomial moments of order $\alpha \geq 1$; i.e., $\mathbb{E}[|S|^p] \leq C^p$ for some $C > 0$ such that $C^p - 1 > 0$.

Assumption 3.1 will be satisfied in cases where the density f is bounded both above and below over the support of f, such as with mass functions on a finite set. However, it may not be satisfied if the log density has no lower bound over the support of f.

Assumption 3.2 will be satisfied when f has full support on $\mathbb{R}^d$ and the log density is unbounded below, such as with a Gaussian distribution.

Assumption 3.3 will be satisfied when f has polynomial tails, such as with a Student-t distribution.

Under each of Assumptions 3.1–3.3, we seek a result on the limiting distribution of the maximum of the surprisals $M_n = \max\{s_1,\dots,s_n\}$. This is provided in Theorem 3.1, with the proof given in the Appendix.

Theorem 3.1. *Let $\boldsymbol{y}_1,\dots,\boldsymbol{y}_n$ be an iid sequence from F, and let $S = -\log f(\boldsymbol{Y})$, where $\boldsymbol{Y} \sim F$. Further, let $M_n = \max\{s_1,\dots,s_n\}$, where $s_i = -\log f(y_i)$.*

1. *If the surprisal satisfies Assumption 3.1, then*

$$\sup_{s:s>0}\left|\Pr\left\{\left|M_n - \mathbb{E}[S]\right| \geq \sqrt{2\nu^2 s} + \sqrt{2\nu^2\log(2n)}\right\} - e^{-s}\right| = o(1).$$

2. *If the surprisal satisfies Assumption 3.2, then*

$$\sup_{s:s>1/b}\left|\Pr\left\{\left|M_n - \mathbb{E}[S]\right| \geq (2b)s + (2b)\log(2n)\right\} - e^{-e^{-s}}\right| = o(1).$$

3. *If the surprisal satisfies Assumption 3.3, then*

$$\sup_{s:s>C}\left|\Pr\left\{\left|M_n - \mathbb{E}[S]\right| \geq (Csn^{1/p})\right\} - e^{-s^{-p}}\right| = o(1).$$

Theorem 3.1 shows that the limiting distribution of M_n falls within the class of extreme value distributions, for each of the three assumptions. If the surprisal has Gaussian like tails, then the maximum surprisal is a reversed Weibull; if it only has an exponential moment, then the maximum surprisal will be Gumbel; if the maximum surprisal only has a polynomial moment, then it is Fréchet. As each of these distributions is a member of the generalized extreme value family, then the upper tail

of the distribution can be approximated by a Generalized Pareto Distribution (Coles 2001, Theorem 4.1).

In particular, even if the assumed distribution F is not correct, provided the tail of the surprisal distribution satisfies one of the three assumptions, we can still fit a GPD to the largest surprisal values, and use this to estimate the tail probabilities.

> **Note**
>
> Add something about what kinds of distributions do not satisfy any of the three assumptions?

4 Experiments

4.1 Standard univariate normal values

Although we need to assume some density function f when computing surprisals, we want the resulting tail probabilities to be robust to a mis-specified distribution. Suppose we generate 1000 observations from a $N(0,1)$ distribution, but use a $t(4)$ distribution to compute the surprisal values. The empirical tail probabilities, and those obtained using the GPD approach, should still give reasonable estimates of the tail probabilities, despite the incorrect distribution being used to compute the surprisals.

To study the effect of the assumed distribution, we compute the surprisal values using both the correct Normal density and an incorrect $t(4)$ density. Then, for each set of surprisal values, we estimate the surprisal tail probabilities in three ways: (a) using the same distribution used to compute the surprisal values; (b) using empirical tail probabilities; and (c) using a GPD fitted to the largest 100 surprisal values.

The various estimates are displayed in the left panel of Figure 1 for observations greater than 2.5, showing how both empirical tail probabilities and the GPD estimates are still accurate, regardless of which distribution was used to compute the surprisals. The solid lines show the results for surprisals computed using the true $N(0,1)$ distribution, while the dashed line shows the results for surprisals computed using the mis-specified $t(4)$ distribution. The estimates computed directly from $t(4)$ ("Assumed distribution") are very inaccurate, but all other estimates lie close to the truth.

In summary, even when we don't know the distribution of the data, we can use some incorrect assumed distribution to compute the surprisals, and then apply either a GPD or the empirical distribution to obtain good estimates of the surprisal probabilities.

The converse situation is shown in the right panel of Figure 1, where the data are generated from a $t(4)$ distribution, and the surprisal values are computed using both the correct $t(4)$ density and an

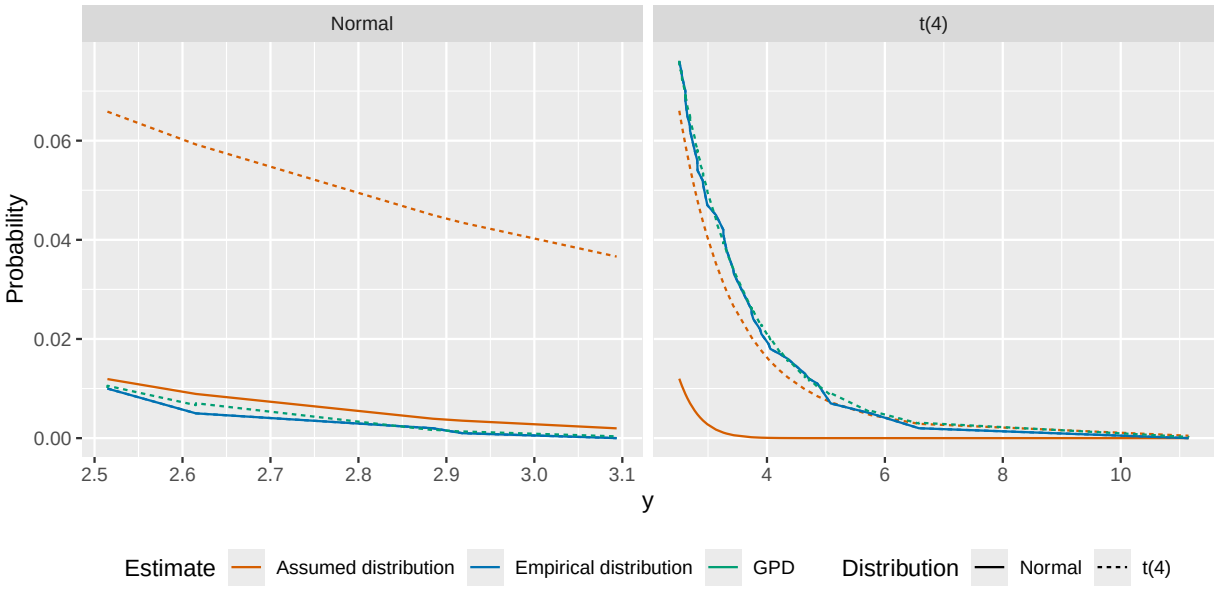


Figure 1: *Left: Observations are computed from a* $\mathbb{N}(,\mathbb{1})$ *distribution. Right: Observations are computed from a* $t(4)$ *distribution. Distribution indicates which distribution was used to compute the surprisal values. Estimate shows how the tail probabilities were computed. This demonstrates that the surprisal probabilities computed using the empirical distribution or under the GPD are still relatively accurate, even when the wrong distribution is used to compute the surprisal values.*

incorrect $N(0,1)$ density. Again, the empirical tail probabilities, and the GPD estimates, are relatively accurate, even when the wrong distribution is used to compute the surprisal values.

4.2 Bivariate gamma values

The quality of the approximation used depends on the amount of data available, so in this next experiment we compare the tail probabilities of the different approaches across sample sizes. Let $\boldsymbol{Y} = (Y_{i,1}, Y_{i,2})'$ where $Y_{i,j}$ are independent Gamma(2,2) random variables. We generate n observations from this bivariate distribution, and explore the quality of the probability estimates as n increases from 100 to 100,000. The surprisal values are computed using the correct distribution, as well as the incorrect bivariate normal distribution with mean $(1,1)'$ and covariance matrix $0.5\boldsymbol{I}$. For each set of surprisal values, we estimate the tail probabilities in two ways: (a) using empirical tail probabilities; and (b) using a GPD fitted to the largest 10% of surprisal values. We repeat this experiment 1000 times for each sample size. The anomaly rate shown is the number of anomalies detected when $\alpha = 0.01$ for different sample sizes, averaged over 1000 replications. The false anomaly rate under the true distribution is, of course, 0.01.

Note that the empirical tail probabilities are identical under both the correct Gamma and incorrect Normal distributions, because the ranking of the surprisals in the tail are unchanged.

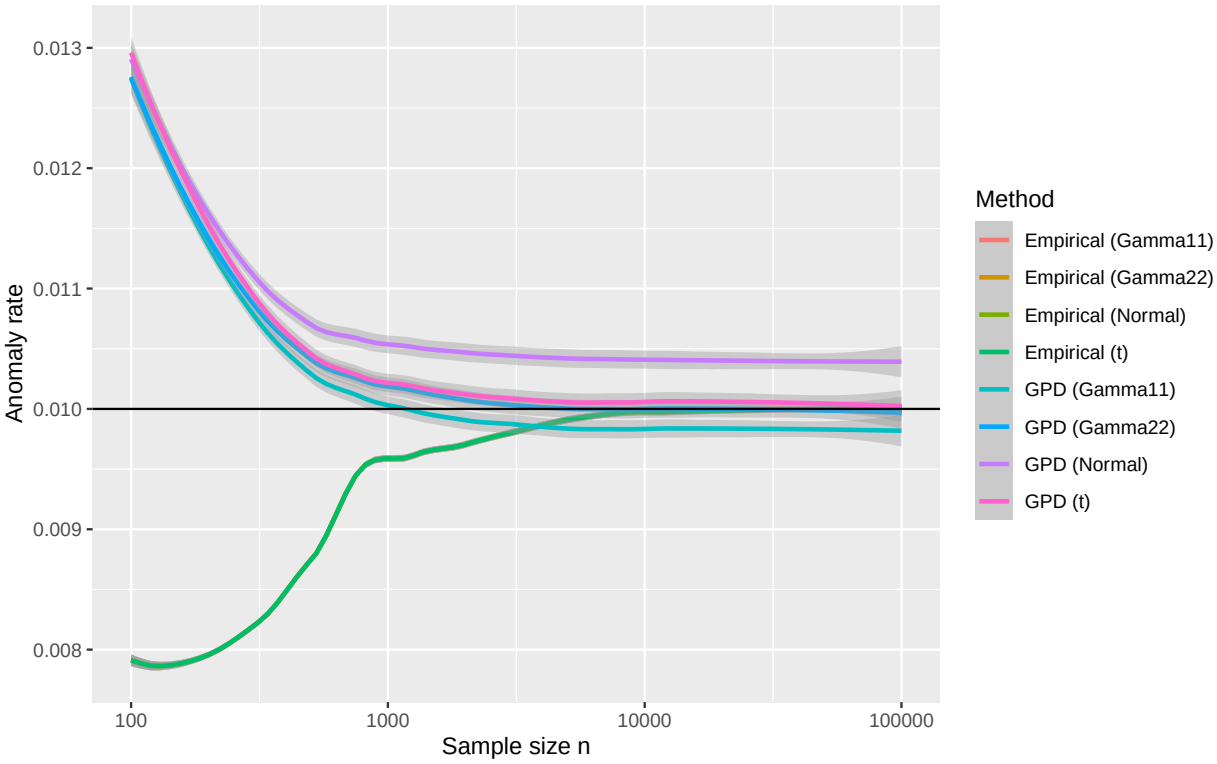


Figure 2: *Estimated false anomaly rate under different approximations when* $\alpha = 0.01$.

> **!** Important
>
> What's going on with the GPD under Gaussian? There seems to be an asymptotic bias.

5 Applications

5.1 French mortality rates

French mortality rates, disaggregated by age (single years) and sex, are available from the Human Mortality Database (2025). We will consider male and female data from 1816 to 1999 over ages 0 to 85, giving 172 separate time series, each of length 184.

Figure 3 shows the data for both sexes and all ages. The mortality rates have improved over time for all ages, especially since 1950. Infant mortality (age 0) is much higher than the rest of childhood, with mortality rates at a minimum at about age 10 in all years. The highest mortality rates are for the oldest age groups. The effect of the two wars are particularly evident in the male mortality rates.

We compute the surprisals using the conditional distribution (1), and then compute the surprisal probabilities under a GPD with $\alpha = 0.01$. With so many observations, there is a high likelihood of many false positives, so we only retain anomalies when at least three age groups are anomalous in the same year, and for the same sex. This results in 289 anomalies out of 31648 observations. Figure 4

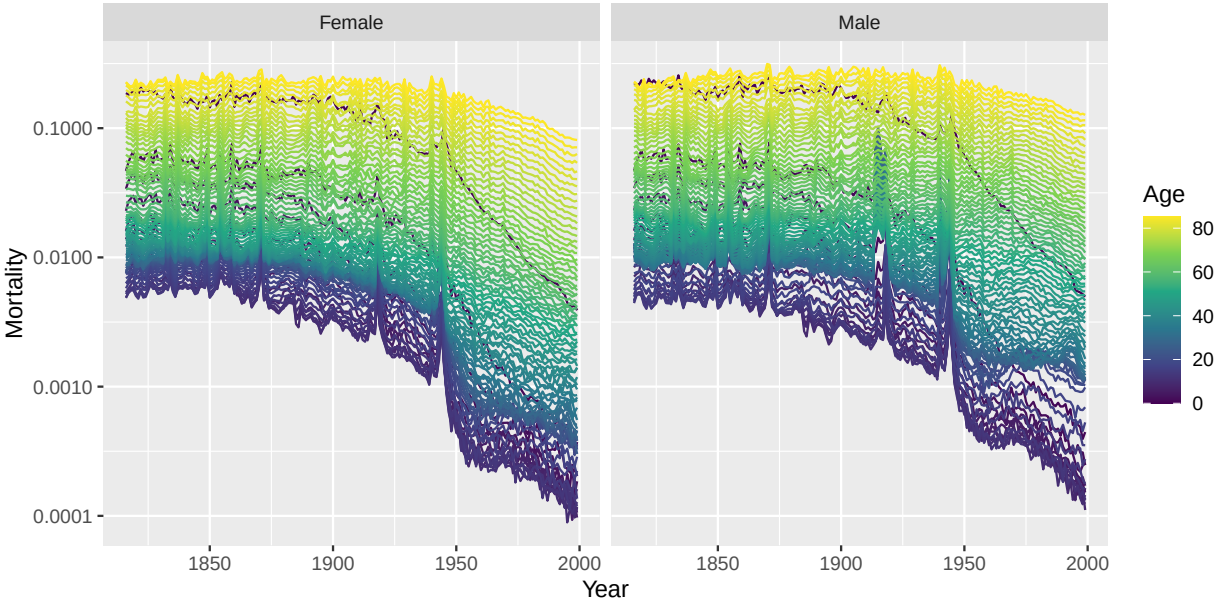


Figure 3: *French mortality rates by sex and age from 1816 to 1999. We use a log scale because the rates are vastly different for different age groups.*

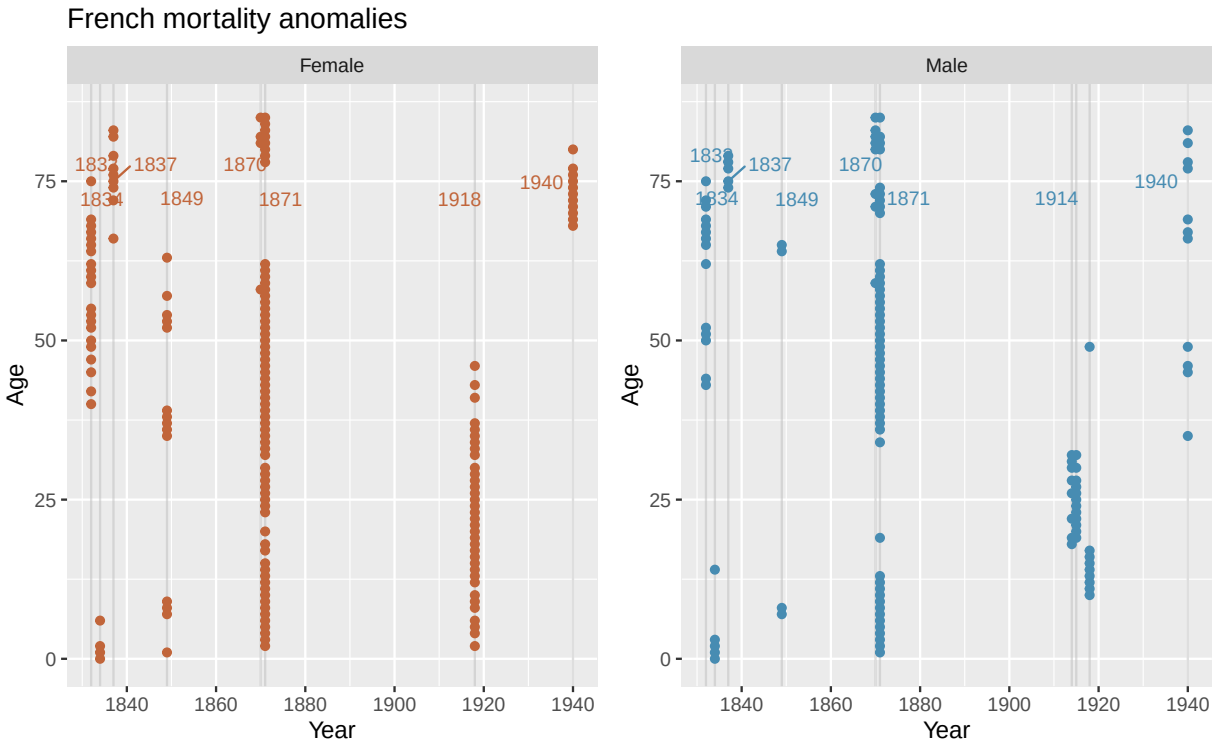


Figure 4: *Anomalies identified in the French mortality data by year and age. Wars and epidemics in French history are revealed.*

shows these anomalies, which coincide with various wars and epidemics that have occurred over French history:

- 1832, 1849: Cholera outbreaks
- 1870: Franco–Prussian war
- 1871: Repression of the 'Commune de Paris'
- 1914-1918: World War I
- 1918: Spanish flu outbreak
- 1940: World War II

In addition, there appears to have been unusually high infant mortality in 1834, and unusually high elderly mortality in 1837. The causes of these is unknown.

5.2 Test cricket not outs

Career batting data for all men and women to have played test cricket between 1834 and 6 October 2025 were obtained from Hyndman (2025). We will investigate if there are some batters who have an unusually high proportion of innings where they have not been dismissed, known as a "not out". The data set contains results from 97649 innings, of which 12695 were not outs. So the overall proportion of not outs is $12695/97649 = 0.130$.

Figure 5 shows the proportion of not outs for each batter as a function of the number of innings they played. The unusual patterns on the left of each plot are due to the discrete nature of the data — both the number of not outs and the number of innings must be integers. There is some overplotting that occurs due to batters having the same numbers of not-outs and innings, which results in the higher color density of the corresponding plotted points. Batters who have played only a few innings tend to have a higher proportion of not outs on average, and a higher variance, than those who have played many innings.

This suggests that we can construct a generalized additive model (Wood 2017) for the number of not outs for each batter as a function of the number of innings they played. It is natural to use a Binomial distribution with a logit link function:

$$\text{NotOuts} \mid \text{Innings} \sim \text{Binomial}(n = \text{Innings},\ p),$$

where p denotes the probability of a batter not being dismissed in an innings,

$$\log(p/(1-p)) = g(\text{Innings}),$$

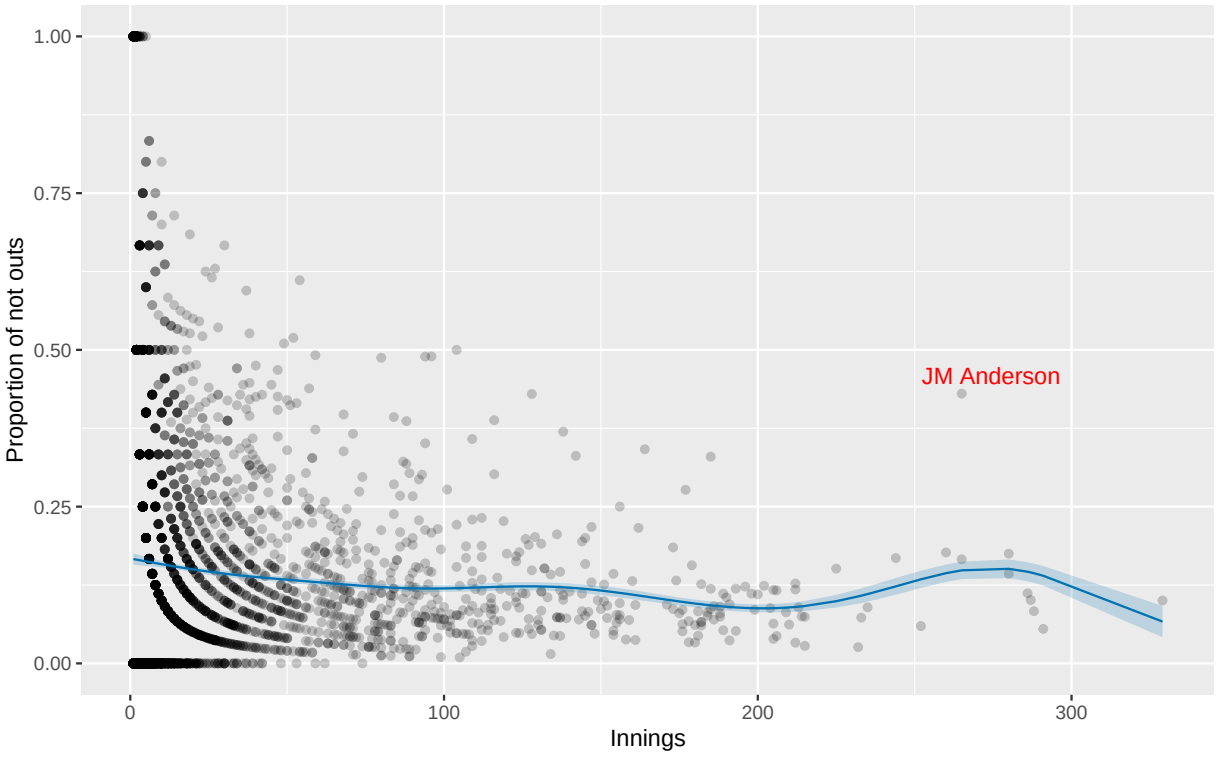


Figure 5: *Proportion of not outs for each batter as a function of the number of innings they played. The blue line and associated 95% confidence interval shows the probability of a batter not being dismissed as a function of the number of innings they have played.*

and $g()$ is a smooth function to be estimated nonparametrically. The resulting estimate, obtained from the mgcv package with penalized regression splines (Wood 2025), is shown in blue in Figure 5.

Now we can use the fitted model to compute the surprisals from the Binomial distribution, and find the most anomalous batters. While the Binomial distribution is probably a good first approximation, it does assume that the probability of a batter being dismissed does not change over their career, which is an unlikely assumption as it does not allow for the development of skill, the value of experience, or the effect of aging. So we will use the GPD approximation to compute the surprisal probabilities.

The most anomalous batters are all "tail-enders" (i.e., not skilled batters) who played for a long time (so they have many innings). Because they batted last, or nearly last, they are more likely to be not out at the end of the team's innings.

The largest surprisal is for English batter Jimmy Anderson, who has had 114 not outs in 265 innings, which is much higher than the expected number of not outs of $265 \times 0.148 = 39.3$. This anomaly is also seen in Figure 5, as being somewhat unusual for that part of the data. Although Jimmy Anderson was not a great batter, he was good at defense, and was able to bat for a long time without being dismissed, leaving the other batter time to score runs.

We have identified an anomaly that is not anomalous in the proportion of not-outs, or in the number of innings, and the difference between the actual proportion and the predicted proportion is not anomalous either compared to some of the other values. However, because we have used a statistical model, we have been able to account for the particular features of this data set, such as the discrete nature of the data, and the changing variance, to identify an observation that is anomalous in the context of the model.

6 Discussion

Software and reproducibility

All results presented here can be reproduced using the code available at https://github.com/robjhyndman/Surprisal_Theory. The analysis was conducted using R version 4.5.1 (R Core Team 2025), with the following R packages: weird (Hyndman 2025), tsibble (Wang et al. 2025), targets (Landau 2025, 2021), ggplot2 (Wickham et al. 2025; Wickham 2016), and other tidyverse (Wickham et al. 2019) packages.

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A Proof of Lemma 2.1

B Proof of Theorem 3.1

We prove each of the three cases in turn.

Proof. Define $\mathbb{P}_n(s) = n^{-1}\sum_{i=1}^{n} 1[S_i \leq s]$. For now, assume that, on the set $s \geq s_\star$, $\widehat{\mathbb{P}}_n(s) = \mathbb{P}_n(s)$ almost surely. If this is satisfied, then

$$\begin{aligned}\Pr\left\{\sup_{s\geq s_\star}\left|\widehat{\mathbb{P}}_n(s) - G(s)\right| > \epsilon\right\} &= \Pr\left\{\sup_{s\geq s_\star}|\mathbb{P}_n(s) - G(s)| > \epsilon\right\} \\ &\leq \Pr\left\{\sup_{s\in\mathbb{S}}|\mathbb{P}_n(s) - G(s)| > \epsilon\right\} \\ &= 2e^{-2n\epsilon^2}, \end{aligned} \tag{1}$$

where the last inequality follows from the Dvoretzky-Kiefer-Wolfowitz theorem, for every $\epsilon > 0$. Taking $\epsilon = \sqrt{\log(2/\alpha)/(2n)}$, we see that

$$\begin{aligned}\Pr\left\{\sup_{s\in\mathbb{S}}|\mathbb{P}_n(s) - G(s)| > \epsilon\right\} &\leq 2e^{-2n\epsilon^2} \\ &= 2e^{-2n\left(\frac{\log(2/\alpha)}{2n}\right)} \\ &= 2e^{-\log(2/\alpha)} \\ &= \alpha.\end{aligned}$$

Hence, with probability at least $1-\alpha$, $\sup_{s\geq s_\star}|\widehat{\mathbb{P}}_n(s) - G(s)| \leq \epsilon$. Fixing an $s \geq s_\star$ and re-arranging terms yields the stated inequality

$$\widehat{\mathbb{P}}_n(s) - \epsilon \leq G(s) \leq \widehat{\mathbb{P}}_n(s) + \epsilon.$$

For the above argument to hold, it must be that, on the set $s \geq s_\star$, $\widehat{\mathbb{P}}_n(s) = \mathbb{P}_n(s)$ almost surely. We now show that this is satisfied if and only if Assumption 2.1 is satisfied.

Case 1: $\Rightarrow$. Take $s \geq s_\star$. Consider the random variable $1[S_i \leq s] = 1$ and note that, by Assumption Assumption 2.1, it must be that $1[h(\widehat{S}_i) \leq h(s)] = 1[S_i \leq s]$ almost surely. However, since $h(\cdot)$ is strictly increasing, it has an inverse that is also strictly increasing, so that (almost surely)

$$1[h(\widehat{S}_i) \leq h(s)] = 1[\widehat{S}_i \leq s].$$

Hence, for $s \geq s_\star$, $\widehat{\mathbb{P}}_n(s) = \mathbb{P}_n(s)$ and (1) is satisfied.

Case 2: $\Leftarrow$. Now, assume that $\widehat{\mathbb{P}}_n(s) = \mathbb{P}_n(s)$ for all $s \geq s_\star$. Then, the sets

$$\mathbb{S} := \{S_1, \ldots, S_n : S_i > s_\star\} \text{ and } \widehat{\mathbb{S}} := \{\widehat{S}_1, \ldots, \widehat{S}_n : \widehat{S}_i > s_\star\}$$

must satisfy $|\mathbb{S}| = |\widehat{\mathbb{S}}|$, since $\widehat{\mathbb{P}}_n(s) = \mathbb{P}_n(s)$. Thus, the values in $\widehat{\mathbb{S}}$ and $\mathbb{S}$ must be the same up to a re-ordering of elements. Hence, there exists a bijection such that $S_i = h(\widehat{S}_i)$ for all $s > s_\star$. □

B.1 S has sub-Gaussian tails

Lemma B.1. *Suppose $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If $S = -\log f(\boldsymbol{Y}) \in \mathbb{R}$, and satisfies, for all $\lambda \in \mathbb{R}$, and some $\nu > 0$,*

$$\mathbb{E} \exp\{\lambda(S - \mathbb{E}[S])\} \leq \exp\{\lambda^2 \nu^2/2\},$$

then, for $s \in \mathbb{R}_+$,

$$\Pr\{M_n - \mathbb{E}[S] \geq s\} \leq 1 - \left(1 - e^{-\frac{1}{2}s^2/\nu^2}\right)^n.$$

Proof. For any $\lambda > 0$,

$$\begin{aligned}\Pr\{S \geq s\} = \Pr\left\{e^{\lambda(S-\mathbb{E}[S])} \geq e^{\lambda(s-\mathbb{E}[S])}\right\} &\leq e^{-\lambda(s-\mu)}\mathbb{E}e^{\lambda(S-\mu)} \\ &\leq e^{-\lambda(s-\mu)}e^{\lambda^2\nu^2/2}.\end{aligned}$$

Now, we can maximize the right-hand side of the above bound in λ, which yields the optimal value $\lambda = (s-\mu)/\{M^2\}$. Plugging this in we have the bound

$$\Pr\{S \geq s\} \leq e^{-\frac{1}{2}(s-\mu)^2/\nu^2}.$$

The stated result now follows by noting that

$$\begin{aligned}\Pr(M_n - \mathbb{E}[S] \geq s) = 1 - \Pr\{M_n \leq s\} &= 1 - \Pr\{S \leq s\}^n \\ &= 1 - (1 - \Pr\{S \geq s\})^n \\ &\leq 1 - \left(1 - e^{-\frac{1}{2}s^2/\nu^2}\right)^n.\end{aligned}$$

□

Corollary B.1. *If Assumption 3.1 is satisfied, then, for all $s > 0$,*

$$\Pr\left(|M_n - \mathbb{E}[S]| \geq \sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(2n)}\right) \leq e^{-s}.$$

Proof. This follows directly from the union bound:

$$\begin{aligned}
\Pr\left(|M_n| \geq \sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(n)}\right) &\leq n \Pr\left\{|S| \geq \sqrt{2}\nu^2 s + \sqrt{2\nu^2 \log(n)}\right\} \\
&\leq 2ne^{-[\sqrt{2\nu^2 s} + \sqrt{2\nu^2 \log(n)}]^2/(2\nu^2)} \\
&= 2ne^{-\frac{1}{2\nu^2}(2s\nu^2) - \frac{2\nu^2 \log(2n)}{2\nu^2} - \frac{\sqrt{2\nu^2 s}\sqrt{2\nu^2 \log(2n)}}{2\nu^2}} \\
&\leq e^{-\sqrt{\log(2n)s} - s} \\
&\leq e^{-s}
\end{aligned}$$

The first result follows from the iid assumption on S and the union bound, while the second from the assumption that S is sub-Gaussian. The remaining inequalities follow from reorganizing terms. □

Corollary B.1 is useful as it clarifies that if the surprisal is sub-Gaussian, then the maximal surprisal will follow a reversed Weibull distribution, and thus is within the class of extreme value distributions.

B.2 S has an exponential moment

Lemma B.2. *Suppose $\mathbf{y}_1, \ldots, \mathbf{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\mathbf{y}_i)$. If Assumption 3.2 is satisfied, then, for $s \in \mathbb{R}_+$,*

$$\Pr(M_n - \mathbb{E}[S] \geq s) \leq 1 - \begin{cases} (1 - e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b \\ (1 - e^{-s/2b})^n & \text{if } s > \nu^2/b \end{cases}.$$

Proof. The result follows a similar argument as for Lemma B.1, up until we maximize the tail bound. In particular, for any $\lambda > 0$,

$$\begin{aligned}
\Pr\{S - \mathbb{E}[S] \geq s\} = \Pr\left\{e^{\lambda(S - \mathbb{E}[S])} \geq e^{\lambda s}\right\} &\leq e^{-\lambda s}\mathbb{E}e^{\lambda(S - \mathbb{E}[S])} \\
&\leq e^{-\lambda s}e^{\lambda^2\nu^2/2}.
\end{aligned}$$

Now, we must consider a constrained optimization to ensure the right-hand side exists. In particular, we know from the proof of Lemma B.1 that the unconstrained optimum is $\lambda = s/\nu^2$, which corresponds to the constrained optimum on the set $0 \leq t < \nu^2/b$. However, when $s \geq \nu^2/b$, the maximum is achieved at $\lambda = 1/b$, and putting this together yields

$$\Pr\{S - \mathbb{E}[S] \geq s\} \leq \begin{cases} e^{-\frac{1}{2}s^2/\nu^2} & \text{if } 0 \leq s \leq \nu^2/b; \\ e^{-s/2b} & \text{if } s > \nu^2/b. \end{cases}$$

The stated result again follows by noting that

$$\begin{aligned}\Pr\{M_n - \mathbb{E}[S] \geq s\} &= 1 - \Pr\{M_n \leq s\} \\ &= 1 - \Pr\{S \leq s\}^n \\ &\leq 1 - \begin{cases} (1 - e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b; \\ (1 - e^{-s/2b})^n & \text{if } s > \nu^2/b. \end{cases}\end{aligned}$$

□

Results similar to Lemma B.2 also hold for $\Pr\{M_n - \mathbb{E}[S] \leq s\}$, and for $\Pr\{|M_n - \mathbb{E}[S]| \geq s\}$ with an additional factor of 2, i.e.,

$$\Pr\{|M_n - \mathbb{E}[S]| \geq s\} \leq 1 - \begin{cases} (1 - 2e^{-\frac{1}{2}s^2/\nu^2})^n & \text{if } 0 \leq s \leq \nu^2/b \\ (1 - 2e^{-s/2b})^n & \text{if } s > \nu^2/b. \end{cases}$$

Corollary B.2. *Under Assumption 3.2, for $s > \nu^2/b$, as $n \to \infty$*

$$\Pr(|M_n - \mathbb{E}[S]| \geq (2b)s + (2b)\log(2n)) \leq 1 - e^{-e^{-s}} + o(1/n).$$

Proof. Apply Assumption 3.2 to see that

$$\begin{aligned}\Pr(|M_n - \mathbb{E}[S]| \geq (2b)s + (2b)\log(2n)) &\leq 1 - \left\{1 - 2e^{-[2bs + 2b\log(2n)]/2b}\right\}^n \\ &= 1 - \left(1 - e^{-s}/n\right)^n \\ &= 1 - e^{e^{-s}} - \left[\left(1 - e^{-s}/n\right)^n - e^{e^{-s}}\right].\end{aligned}$$

The result now follows by noting that, for all $s > 0$,

$$\left[\left(1 - e^{-s}/n\right)^n - e^{e^{-s}}\right] = o(1) \leq 0.$$

□

Corollary B.2 is useful as it shows that if the surprisal has a sub-exponential tail, then the resulting maximum surprisal tail probability converges to that of an extreme value distribution. In particular, we have that Corollary B.2 implies

$$\sup_{s:s>\nu^2/b} \left|\Pr\left\{|M_n - \mu| \geq \left[2bs + 2b\log(2n)\right]\right\} - \left(1 - e^{e^{-s}}\right)\right| = o(1).$$

To see that this implies that $M_n - \mathbb{E}[S]$, appropriately standardized, converges to an extreme value distribution, we first recall that

$$\Pr\{X_n > x\} - \Pr\{X > x\} = 1 - \Pr\{X_n \le x\} - [1 - \Pr\{X \le x\}] = \Pr\{X \le x\} - \Pr\{X_n \le x\}.$$

Hence,

$$\sup_{x \in \mathbb{X}} |\Pr\{X_n \le x\} - \Pr\{X \le x\}| = \sup_{x \in \mathbb{X}} |\Pr\{X_n > x\} - \Pr\{X > x\}|,$$

so that when X_n and X are both continuous we see that Corollary B.2 implies that the CDF of $(M_n - \mathbb{E}[S] - 2b\log(2n))/2b$ converges to that of the Type-1 extreme value distribution (i.e., Gumbel distribution); i.e., using Corollary B.2 and the above we have that

$$\sup_{s : s > v^2/b} \left| \Pr\left(|M_n - \mu| \le [2bs + 2b\log(2n)]\right\} - e^{e^{-s}} \right| = o(1).$$

This is one notion of convergence in distribution.

B.3 S has a polynomial moment

Suppose $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_n$ are iid from a density f, and let $M_n = \max\{s_1, \ldots, s_n\}$ where $s_i = -\log f(\boldsymbol{y}_i)$. If Assumption 3.3 is satisfied, then, for $s > 0$ and such that $s > C > 0$,

$$\Pr\left(|M_n - \mathbb{E}[S]| \ge s\right) \le 1 - (1 - C^p/s^p)^n.$$

Hence,

$$\sup_{s : s > C} \left| \Pr\left(|M_n - \mathbb{E}[S]| \le (s \cdot n^{1/p})\right) - e^{-s^{-p}} \right| = o(1).$$

Proof. By Markov's inequality,

$$\Pr\{|S - \mathbb{E}[S]| \ge s\} \le \frac{\mathbb{E}|S|^p}{s^p} \le \frac{C^p}{s^p}.$$

To derive the first stated result, again note that

$$\begin{aligned}
\Pr\{|M_n - \mathbb{E}[S]| \ge s\} &= 1 - \Pr\{|M_n - \mathbb{E}[S]| \le s\} \\
&= 1 - \Pr\{|S - \mathbb{E}[S]| \le s\}^n \\
&= 1 - (1 - \Pr\{|S - \mathbb{E}[S]| \ge s\})^n \\
&\le 1 - (1 - C^p/s^p)^n.
\end{aligned}$$

To derive the second stated result, apply the first result with $t = sn^{1/p}C > 1$, which clearly holds for some n large enough, to see that

$$\Pr\{|M_n - \mathbb{E}[S]| \geq t\} \leq 1 - \left(1 - \frac{C^p}{(Csn^{1/p})^p}\right)^n = 1 - \left(1 - s^{-p}/n\right)^n.$$

Now, recall that, for $x \geq 0$,

$$\lim_{n\to\infty} (1 - x/n)^n = e^{-x}.$$

Since $x \mapsto e^{-x}$ is bounded for all $x \geq 0$, it follows that

$$\sup_{s:s>C} \left|\Pr\{|M_n - \mathbb{E}[S]| \geq t\} - e^{-s^{-p}}\right| = o(1).$$

□